

a functional mapping to an RV. We discuss functions (transformations) of random variables in Section 2.3.

- Just as we can have vectors of numbers, we can also have vectors of RVs, and so we discuss groups of random variables and random vectors in Section 2.4.
- Just as we can have scalar functions of time, we can also have RVs that are functions of time, and so we discuss RVs that change with time (stochastic processes) in Section 2.5.
- Stochastic processes can be divided into two categories: white noise and colored noise, and we discuss these concepts in Section 2.6.

We conclude in Section 2.7 with a high-level discussion of how to write a computer simulation of a noise process.

This chapter is only a brief introduction and review of probability and stochastic processes, and more detail can be found in many other books on the subject, such as [Pap02, Pee01].

2.1 PROBABILITY

How shall we define the concept of probability? Suppose we run an experiment a certain number of times. Sometimes event A occurs and sometimes it does not. For instance, our experiment may be rolling a six-sided die. Event A may be defined as the number 4 showing up on the top surface of the die after we roll the die. Common sense tells us that the probability of event A occurring is $1/6$. Likewise, we would expect that if we run our experiment many times, then we would see the number 1 appearing about $1/6$ of the time. This intuitive explanation forms the basis for our formal description of the concept of probability. We define the probability of event A as

$$P(A) = \frac{\text{Number of times } A \text{ occurs}}{\text{Total number of outcomes}} \quad (2.1)$$

This commonsense understanding of probability is called the relative frequency definition. A more formal and mathematically rigorous definition of probability can be obtained using set theory [Bil95, Nel87], which was pioneered by Andrey Kolomogorov in the 1930s. But for our purposes, the relative frequency definition is adequate.

In general, we know that there are n -choose- k different ways of selecting k objects from a total of n objects (assuming that the order of the objects does not matter), where n -choose- k is denoted and defined as

$$\binom{n}{k} = \frac{n!}{(n-k)!k!} \quad (2.2)$$

For instance, suppose we have a penny (P), nickel (N), dime (D), and quarter (Q). How many distinct subsets of three coins can we pick from that set? We can pick PND, PNQ, PDQ, or NDQ, for a total of four possible subsets. This is equal to 4-choose-3.

■ EXAMPLE 2.1

What is the probability of being dealt four of a kind¹ in poker? The total number of possible poker hands can be computed as the total number of subsets of size five that can be picked from a deck of 52 cards. The total number of possible hands is 52-choose-5 = 2,598,960. Out of all those hands, there are 48 possible hands containing four aces, 48 possible hands containing four kings, and so on. So there are a total of 13×48 hands containing four of a kind. Therefore the probability of being dealt four of a kind is

$$\begin{aligned} P(A) &= \frac{(13)(48)}{2,598,960} \\ &= 1/4165 \\ &\approx 0.024\% \end{aligned} \tag{2.3}$$

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The conditional probability of event A given event B can be defined if the probability of B is nonzero. The conditional probability of A given B is defined as

$$P(A|B) = \frac{P(A, B)}{P(B)} \tag{2.4}$$

$P(A|B)$ is the conditional probability of A given B , that is, the probability that A occurs given the fact that B occurred. $P(A, B)$ is the joint probability of A and B , that is, the probability that events A and B both occur. The probability of a single event [for instance, $P(A)$ or $P(B)$] is called an *a priori* probability because it applies to the probability of an event apart from any previously known information. A conditional probability [for instance, $P(A|B)$] is called an *a posteriori* probability because it applies to a probability given the fact that some information about a possibly related event is already known.

For example, suppose that A is the appearance of a 4 on a die, and B is the appearance of an even number on a die. $P(A) = 1/6$. But if we know that the die has an even number on it, then $P(A) = 1/3$ (since the even number could be either a 2, 4, or 6). This example is intuitive, but we can also obtain the answer using Equation (2.4). $P(A, B)$ is the probability that both A occurs (we roll a 4) and B occurs (we roll an even number), so $P(A, B) = 1/6$. So Equation (2.4) gives

$$\begin{aligned} P(A|B) &= \frac{1/6}{1/2} \\ &= 1/3 \end{aligned} \tag{2.5}$$

The *a priori* probability of A is $1/6$. But the *a posteriori* probability of A given B is $1/3$.

■ EXAMPLE 2.2

Consider the eight shapes in Figure 2.1. We have three circles and five squares, so $P(\text{circle}) = 3/8$. Only one of the shapes is a gray circle, so $P(\text{gray, circle})$

¹Once I was dealt four sevens while playing poker with some friends (unfortunately, I was not playing for money at the time). I don't expect to see it again in my lifetime.

$= 1/8$. Of the three circles, only one is gray, so $P(\text{gray} | \text{circle}) = 1/3$. This last probability can be computed using Equation (2.4) as

$$\begin{aligned} P(\text{gray}|\text{circle}) &= \frac{P(\text{gray, circle})}{P(\text{circle})} \\ &= \frac{1/8}{3/8} \\ &= 1/3 \end{aligned} \tag{2.6}$$

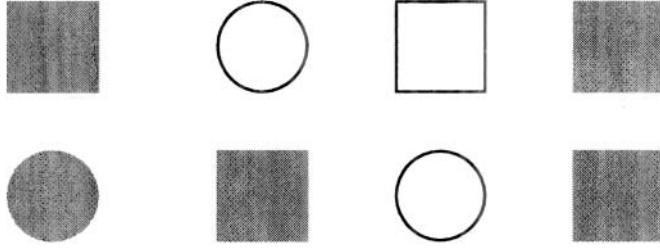


Figure 2.1 Some shapes for illustrating probability and Bayes' Rule.

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Note that we can use Equation (2.4) to write $P(B|A) = P(A, B)/P(A)$. We can solve both this equation and Equation (2.4) for $P(A, B)$ and equate the two expressions for $P(A, B)$ to obtain Bayes' Rule.

$$P(A|B)P(B) = P(B|A)P(A) \tag{2.7}$$

Bayes' Rule is often written by rearranging the above equation to obtain

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \tag{2.8}$$

As an example, consider Figure 2.1. The probability of picking a gray shape given the fact that the shape is a circle can be computed from Bayes' Rule as

$$\begin{aligned} P(\text{gray}|\text{circle}) &= \frac{P(\text{circle}|\text{gray})P(\text{gray})}{P(\text{circle})} \\ &= \frac{(1/5)(5/8)}{3/8} \\ &= 1/3 \end{aligned} \tag{2.9}$$

We say that two events are independent if the occurrence of one event has no effect on the probability of the occurrence of the other event. For example, if A is the appearance of a 4 after rolling a die, and B is the appearance of a 3 after rolling *another* die, then A and B are independent. Mathematically, independence of A and B can be expressed several different ways. For example, we can write

$$\begin{aligned} P(A, B) &= P(A)P(B) \\ P(A|B) &= P(A) \\ P(B|A) &= P(B) \end{aligned} \tag{2.10}$$